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Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

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Abstract

With climate change, there's an increasing need for a higher-resolution wildfire data for better monitoring and risk assessment. GEO satellites have high-temporal resolution, but lacks spatial resolution for a detailed risk assessment. While LEO satellites provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing downscaling models have demonstrated their effectiveness in producing high-resolution remote sensing data. But many still fail to capture the fine details. This project aims to achieve a high-resolution spatial downscaling of wildfire data by integrating additional physical covariates to the model to mitigate implausible outputs. The project explores direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-resolution wildfire data. The project compares effectiveness of hybrid attention U-Net (HAUNet) as wildfire downscaling model.

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1 Introduction

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). This causes economic losses for the insurance industry. For instance, The January 2025 Los Angeles County wildfires set an all-time record, causing $40 billion in insured losses alone from a single event (Dwyer and Nam, 2026). This project, conducted with Artio, aims to address this gap. By providing highly accurate risk assessment models, this research seeks to give better confidence and enable better risk assessment.

High-spatial and high-temporal resolution wildfire data is favourable in risk assessment. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020a). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate error caused by the large resolution difference. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net.

2 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns in remote sensing data. Recently, Deep learning methods have shown to outperform these approaches (Sdraka et al., 2022). CNN and U-Net architectures have been widely used for their ability to capture spatial features. While GAN-based methods have been used where sharpness is prioritised over pixel-wise accuracy. More recently, transformer-based models use long-range attention to capture dependencies beyond a convolution's receptive field (J. Wang et al., 2025), and diffusion-based approaches have showed strong generative ability (Y. Wang et al., 2023).

However, many downscaling architectures are optimised for texture-rich images. They are not well-suited to the distinct structure of active wildfire data where the target is very sparse. This creates two challenges: severe class imbalance, which biases pixel-wise reconstruction losses and convolutional decoders (Qi et al., 2026); and an ill-posed one-to-many mapping, where a single low-resolution pixel have many possible high-resolution configurations (Z. Wang et al., 2020b). Architectures are needed that can both preserve fire signal against a dominant background and recover coherent fire-front structure.

Models that downscale across large reoslution gaps with only the coarse fire input have to rely on the learned spatial patterns from the coarse data. While wildfires are influenced by physical factors such as terrain and landcover (Sullivan, 2017), generic super-resolution models omit this information. Providing HR covariates provides additional context and constraints the model can learn from, which can help mitigate the ill-posed nature of the downscaling problem.

Existing literature is dominated by direct, single-model downscaling across the full resolution gap. While multi-stage or task-decoupled downscaling methods exist (such as Li et al., 2023 that

combines spatial mixing with residual compensation, or decoupling classification and regression H. Wang et al., 2026), no prior work researches cascaded DL downscaling, particularly for wildfire applications. Specifically, no prior work evaluates a multi-stage physical downscaling pipeline that sequentially trains model architectures trained with independent satellite target datasets at intermediate spatial resolutions (e.g. training Stage 1 to map 2km → 375m using VIIRS target grids, and Stage 2 to map 375m → 30m using Landsat/Sentinel target grids). Existing research also has not explored the trade-off between the reduced per-stage resolution gap and the accumulation of error across stages.

To address these gaps, the objectives of this project are:

1. Develop a domain-specific, multi-stage physical downscaling framework tailored to the extreme target sparsity and physical constraints of active wildfires.

2. Compare the effectiveness of a Hybrid Attention U-Net (HAUNet) against a baseline convolutional autoencoder in preserving localised active fire signals and suppressing background reconstruction noise.

3. Evaluate the performance and trade-offs of a multi-stage physical pipeline against direct downscaling, assessing whether cascaded approach improves overall active fire spatial accuracy.

3 Data and Materials

Table 1 shows the data sources used for this project. The specifications for the VIIRS, GOES, and Landsat data are outlined in Table 2.

The GOES-16/17 data were obtained via public Google Cloud Platform (GCP) buckets (NOAA, 2017, 2018). DEM, and ESA WorldCover are obtained via Google Earth Engine (GEE) Python API (Gorelick et al., 2017) and exported directly to Google Cloud Storage (GCS). The Landsat-8 fire masks are derived by de Almeida Pereira et al., 2021 and obtained from the author's public dataset (hosted on Google Drive) and re-uploaded to GCS for pipeline consistency. VIIRS fire detections are obtained from NASA FIRMS archive. Manifests, split assignments and model checkpoints are also stored in GCS.

There are 1,2504 and 1,202 total VIIRS and Landsat patches respectively. The assymetry in sample counts is due to different data sources, Landsat from a pre-existing external dataset, VIIRS generated directly from a larger available detection archive. This gives Stage 1 (GOES → VIIRS) of the cascaded pipeline a larger, more reliable training set, since Stage 1 errors propagate directly into Stage 2's input. But the final 30m downscaling is trained on the same Landsat dataset.

The data was processed so that all regions of interest (ROIs) are dynamically reprojected to their localised UTM zone based on centroid coordinates. Detections from the NASA VIIRS S-NPP archive were filtered for "nominal" and "high" confidence points. Detections are clustered into 7680 × 7680 m grid cells within a 15-minute temporal window of peak activity. VIIRS grid cell pixels record the peak Fire Radiative Power (FRP). Due to extreme class imbalance of the target fire datasets (mean fraction of positive target pixels per patch are 0.85% for VIIRS, 0.06% for Landsat), stratified sampling was applied across temporal extent and fire pixel count bins during data splitting. The closest temporal GOES-16/17 scan within a ±60-minute guard window was queried.

GOES-16/17 channels 7 (3.9 μm, shortwave IR) and 14 (11.2 μm, longwave IR) are used as they directly correspond to the spectral bands for fire-detection for their sensitivity to fire radiance. This is similar spectral logic in VIIRS's own active-fire algorithm, which similarly relies on a shortwave (I4, 3.74 μm) and longwave (I5, 11.45 μm) channel pair. (Schmidt et al., 2013)

In addition to the native 3.9 μm (Ch. 7) and 11.2 μm (Ch. 14) GOES brightness temperatures, a Thermal Fire Index (TFI) is computed as additional input channel:

$$TFI = Ch.7 - Ch.14 \tag{1}$$

Explicitly providing the difference reduces the burden on early network layers to discover a known, physically meaningful relationship, aiming to save model capacity and improving convergence.

The dataset is split into training/validation/testing (80/10/10). This split utilises stratified random sampling across four quantile bins based on the active fire pixel count per patch for balanced representation of fire sizes. Weighted random sampling is applied during training to address class imbalance, where sample weights w_i for each patch i were assigned based on the count of positive fire pixels n_i:

$$w_i = \max(n_i, 1) \tag{2}$$

where n_i is extracted from the patch metadata manifests.

Inputs are scaled prior to model ingestion. The GOES and DEM channels undergo Z-score standardisation using mean and standard deviation parameters precomputed exclusively from the training split. DEM covariates were normalised by dividing by 1000 (converting to km). The landcover data is also one-hot encoded into 11 channels, each representing a landcover class.

For the targets, the continuous VIIRS FRP values are binarised into active fire masks. Landsat targets are already binarised. Depending on the experimental configuration, these are transformed using either spatial dilation via max pooling operation.

Spatial dilation was applied using a 3×3 max-pooling kernel for the VIIRS branch and a 7×7 for the Landsat branch. Landsat targets are much sparser than the VIIRS targets, so a larger dilation kernel was used to provide a comparably denser training signal.

The Landsat branch has a limited number of samples (1,202 total), and given the extreme class imbalance in fire pixel prevalence, training samples were augmented via independent horizontal and vertical flips (each applied with probability 0.5), applied to the input and target. This increases the effective diversity of the training set. No augmentation was applied to validation or test splits.

4 Methods

4.1 HAUNet (Hybrid Attention U-Net)

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses hybrid attention to capture both local fine detail and broader contextual structure across multiple scales. HAUNet was selected for its ability to capture local extreme hotspots in near-zero background, which standard U-Nets might smooth out.

4.1.1 Hybrid Attention Mechanisms

- **Channel Attention Module (CEM):** Computes channel dependencies with $O(C^2)$ complexity. Given feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions followed by 3×3 depthwise convolutions. The channel attention map $\mathbf{A}_c$ is derived as:

$$\mathbf{A}_c = \text{Softmax}\left(\text{ReLU}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\alpha}\right)\right)\mathbf{V} \tag{3}$$

  where α is a learnable scaling parameter.

Table 1: Data Sources and Specifications

Data Source	**Res.**	**Bands Used**	**Role / Context**
Landsat-8 Fire Masks	30 m	Active Fire Mask de Almeida Pereira et al., 2021	Target / ground truth; binary mask (256×256 px)
VIIRS S-NPP	375 m	I4 (3.74 μm, Active Fire)	Intermediate target / cascaded-pipeline input (FRP, NASA FIRMS)
GOES-16/17 (ABI)	2000 m	Ch. 7 (3.9 μm), Ch. 14 (11.2 μm)	Coarse fire input; brightness temperature, high temporal frequency
Copernicus DEM	30 m	DEM	Static covariate; terrain effects on fire spread
ESA WorldCover	10 m	Landcover (10 classes)	Static covariate; land cover map

Table 2: Data Pipeline Configuration Specifications

Parameter	**Value**	**Description**
VIIRS Geographic Bounds	$[-170.0, -30.0]$	Longitude limits for the Americas
VIIRS Date Range	2020-01-01 – 2020-12-31	Temporal query window for VIIRS detections
Landsat Geographic Bounds	N. & S. America	Locations of interest for GOES detections
Landsat Date Range	Aug–Sep 2020	Temporal window for Landsat detections de Almeida Pereira et al., 2021
Min. Landsat Active Pixels	10	Minimum active fire pixels to retain a patch
GOES Temporal Guard	60 min	Max. gap between target time and GOES scan
GOES Native Scale	2000 m	Native GOES ABI grid spacing
Landsat Grid Scale	30 m	Resolution, Landsat branch
VIIRS Grid Scale	375 m	Resolution, VIIRS active fire detections
Landsat Dimensions	256×256 px	Target patch size
VIIRS Dimensions	20×20 px	Target size (7680/375)
GOES Dimensions	4×4 px	Target size (7680/2000)

- **Spatial-Channel Attention (S_CEM):** Parallel spatial attention branch operating along transposed dimensions $\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s \in \mathbb{R}^{B \times N_{head} \times HW \times C}$. The full spatial attention step scales quadratically with spatial resolution as $O((HW)^2)$.

4.1.2 Full-Resolution Covariate Pyramid Branch

Because full spatial attention in S_CEM imposes an $O((HW)^2)$ memory footprint that causes out-of-memory errors on full-resolution inputs, the inputs (256×256) are downsampled by factor $D = 4$ (i.e. $256/4 = 64$) before passing to the model.

The parallel Covariate Pyramid Encoder was introduced because downsampling covariates alongside the fire input discards the HR detail at the target 30m scale. Processing covariates at full resolution in a separate branch, then progressively fusing into the main decoder path. This branch processes full-resolution covariate features using Windowed Spatial-Channel Attention Blocks (Windowed-S_CEM) (Liu et al., 2021) for computational efficiency.

For a feature map partitioned into non-overlapping local windows of size $P \times P$ ($P = 8$), Windowed-S_CEM computes spatial attention independently within each window. The window is shifted sequentially with shift size $P/2$. This keeps the spatial attention complexity to $O(P^4)$ per window, independent of overall spatial input dimensions.

4.1.3 Multi-Scale Refinement and Feature Fusion

The main encoder-decoder path produces low-resolution feature representations (64×64). The final refinement stage produces the final high-resolution output. At each stage:

1. Low-resolution features are upsampled via bilinear interpolation followed by 3×3 convolutions.

2. Upsampled features are concatenated with matching scale representations extracted from the Covariate Pyramid Encoder.

3. A 3×3 fusion convolution maps concatenated features back to channel dimension.

The final layer of the HAUNet refinement network outputs prediction at the resolution of the full covariate grid (256×256). Operating the decoder at full resolution preserves fine-grained spatial feature representations across skip connections and covariate channels. This is same for Stage 1 (GOES $\rightarrow$ VIIRS) model output for cascaded pipeline. Rather than constraining the output's resolution to match the coarse VIIRS target grid (20×20), HAUNet outputs high-resolution spatial feature maps. During metric evaluatation of the Stage 1 model, the outputs are pooled to VIIRS resolution to match VIIRS target.

4.1.4 Architecture Selection Rationale

HAUNet was selected as one of the architectures for the following reasons:

1. It uses dual attention: channel attention to capture global information across feature channels and spatial attention to capture spatial relationships. This is useful for wildfire data, where the fire signal is sparse and localised, but the physical context (e.g. terrain, landcover) is important for accurate downscaling.

2. Its CIM used as skip connections. Skip connections used in conventional U-Net concatenate features at matching spatial resolution, but this assume they are semantically comparable. CIM addresses this by explicitly bridging semantic and resolution gaps between encoder outputs at different scales prior to fusion. This helps to better integrate and learn representation at different scales, which is useful for super-resolution tasks like this (downscaling by 66.7×).

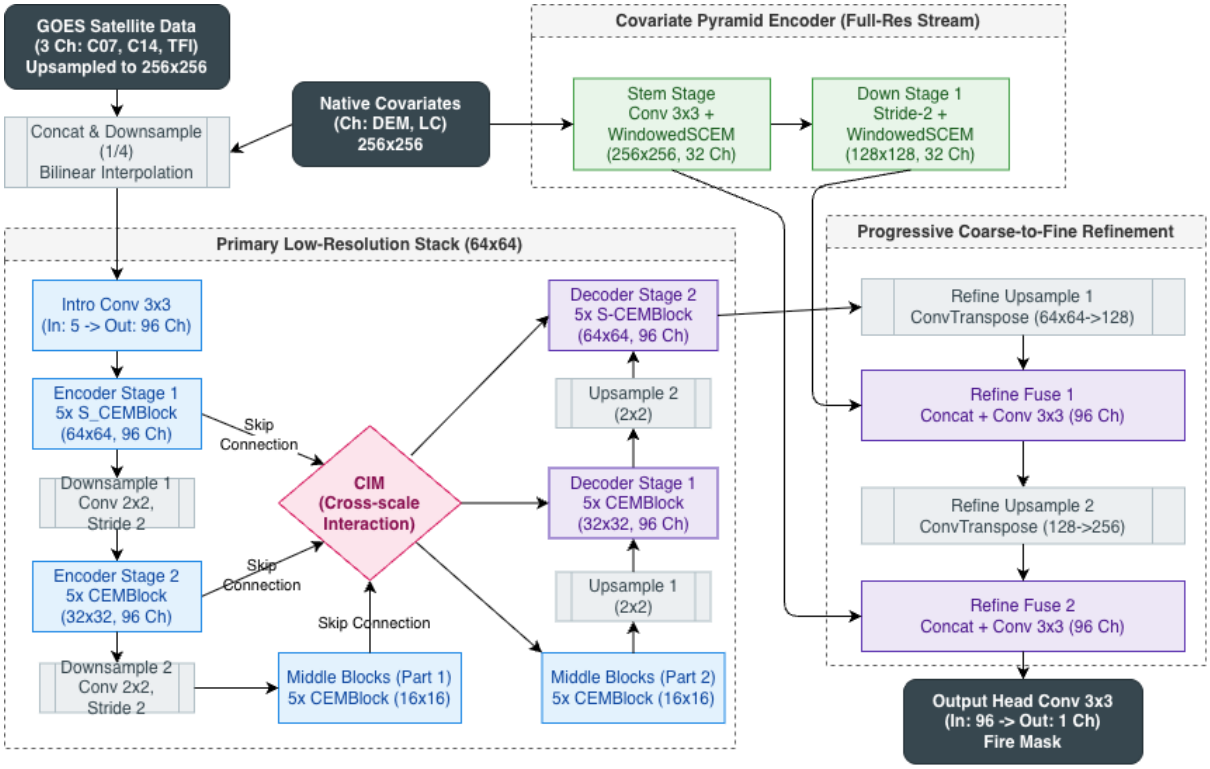


Figure 1: Architecture overview of the wildfire-adapted HAUNet. The full-resolution stream processes native covariates, which are progressively fused via refinement blocks with the low-resolution stack after CIM.

3. Both the autoencoder baseline (4.2) and HAUNet share the similar underlying encoder-decoder-with-skip-connections structure. This allows the comparison between the two models to directly isolate the contribution of HAUNet's attention mechanisms and skip-connection design.

4.2 Autoencoder Baseline

The baseline architecture follows the autoencoder proposed by Badhan et al., 2024 to downscale fire data from GOES to VIIRS resolution. Here, it is adapted to this project to concatenate the covariates with the fire data inputs, and outputs are fire mask segmentation rather than continuous BT values. The autoencoder follows a standard U-Net-style structure, shown in Figure 2.

This baseline is trained under identical conditions to HAUNet (same loss, optimiser, data splits, and covariate set) so that any performance gap are attributable to architectural differences. It establishes a lower bound against which HAUNet's improvement can be quantified, and isolating whether performance gains is from attention and full-resolution covariate processing, or would be achievable with a standard convolutional model.

4.3 Loss Functions

4.3.1 Focal Dice Loss

Active fire pixels represent a small fraction of each patch, resulting in extreme class imbalance. To address this, Focal Dice Loss was used. It combines Focal Loss (which down-weights background pixels via $(1 - p_t)^\gamma$) with Dice Loss, which optimises for spatial region overlap:

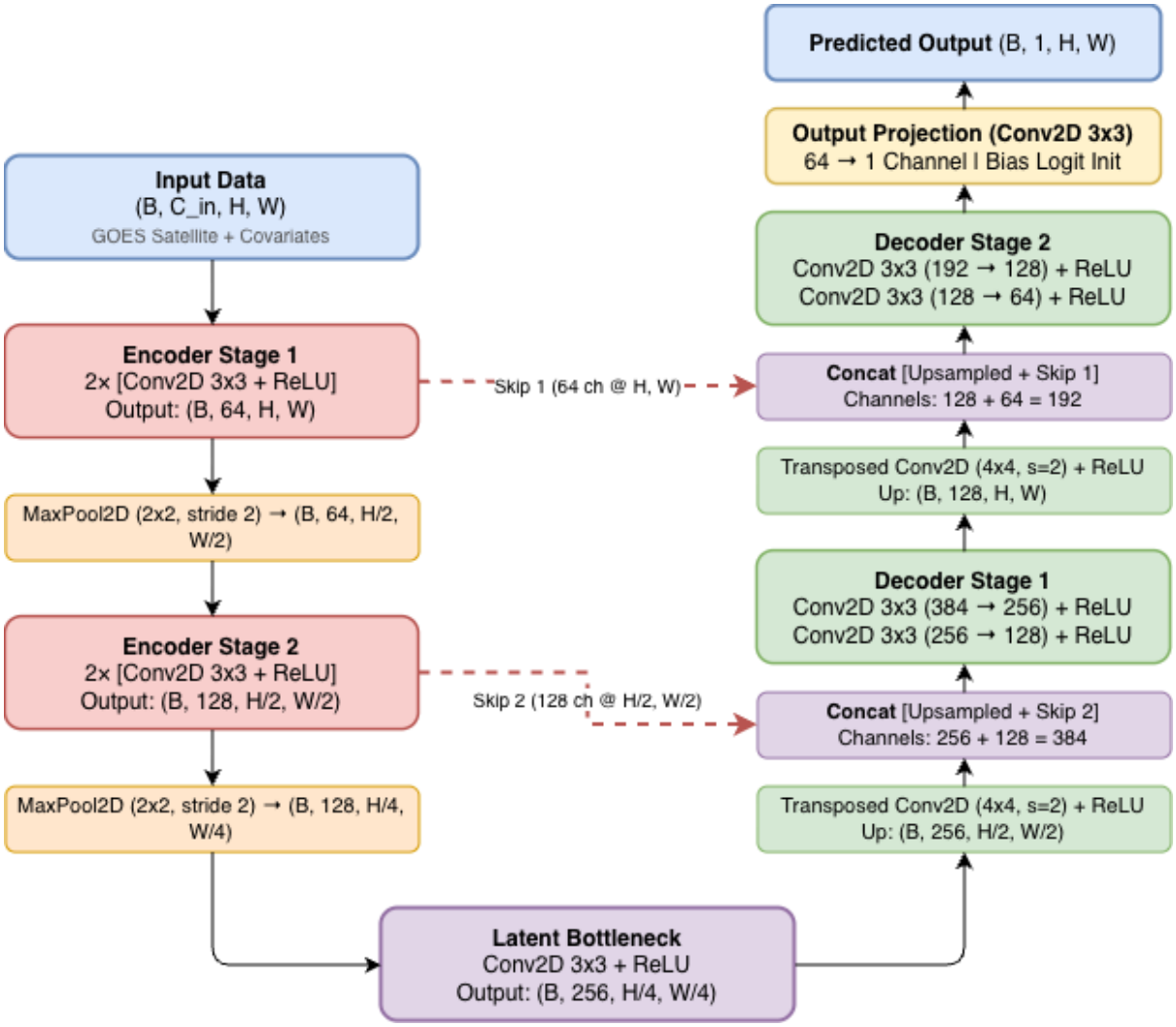


Figure 2: Architecture overview of the autoencoder baseline based on Badhan et al., 2024.

$$\mathcal{L}_{\text{FD}} = w_f \cdot \mathcal{L}_{\text{focal}} + w_d \cdot \mathcal{L}_{\text{dice}} \tag{4}$$

where w_f and w_d are weights of the losses.

Focal Loss (Lin et al., 2018) applies $(1 - p_t)^\gamma$ to focus training on sparse positives:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{5}$$

where α_t is the balancing factor, p_t is the predicted probability for the ground-truth class, and $p_{i,j} = P(y_{i,j} = 1)$ is the estimated probability for the positive class. For a ground-truth label $y_{i,j} \in \{0, 1\}$:

$$p_t = \begin{cases} p_{i,j} & \text{if } y_{i,j} = 1 \\ 1 - p_{i,j} & \text{if } y_{i,j} = 0 \end{cases}, \quad \alpha_t = \begin{cases} \alpha & \text{if } y_{i,j} = 1 \\ 1 - \alpha & \text{if } y_{i,j} = 0 \end{cases} \tag{6}$$

$\alpha \in [0, 1]$ balances positive and negative pixel contributions, and $\gamma \geq 0$ controls the rate at which pixels are down-weighted.

Dice Loss (Milletari et al., 2016) tries to maximise the spatial overlap:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{i,j} p_{i,j} y_{i,j} + \epsilon}{\sum_{i,j} p_{i,j} + \sum_{i,j} y_{i,j} + \epsilon} \tag{7}$$

where ϵ is a smoothing term preventing division by zero.

4.4 Computational Environment

Model development and training were conducted on a GCP Vertex AI Workbench instance equipped with an NVIDIA L4 GPU, 23.7 GB VRAM, 32 GB RAM, 8 vCPUs. Data generation and preprocessing were run as separate CPU-only jobs querying GCS directly.

4.5 Experimental Design

HAUNet was assessed under the following ablations, in addition to the primary direct-vs-cascaded pipeline comparison:

1. with and without the Covariate Pyramid Encoder.
2. with and without DEM as an input covariate.
3. with and without land cover as an input covariate.
4. with and without target dilation (to assess the effect of target representation on model performance).

In the cascaded pipeline, Stage 1 (GOES to VIIRS) is trained on GOES/VIIRS pairs to predict a VIIRS-resolution (375m) fire mask from the GOES input. Stage 2 (VIIRS to Landsat) is intended to further downscale this to Landsat resolution (30m). However, VIIRS and Landsat overpasses times do not overlap, so matching VIIRS/Landsat pairs are not directly available in the same way GOES/VIIRS pairs are. To address this temporal mismatch, Stage 2 is instead trained using the trained Stage 1 model's predictions on GOES/Landsat dataset as inputs, rather than ground-truth VIIRS observations. For each GOES/Landsat pair, the Stage 1 model produces a predicted VIIRS-resolution fire mask from the temporally matched GOES input, and this prediction is paired with the corresponding Landsat ground truth to train Stage 2. This ensures both stages of the cascaded pipeline operate on inputs available at inference time (i.e. Stage 2 never has access to real VIIRS observations at test time either), avoiding a train/inference mismatch that would occur if Stage 2 were trained on ground-truth VIIRS data unavailable in a genuinely cascaded operational setting.

Dilation was investigated for potentially mitigating the severe class imbalance as it may help the model to learn fire regions more effectively. However, dilation can blur the boundaries of fire regions, potentially leading to less precise predictions. The effectiveness of this approach was evaluated through experiments comparing models trained with and without target dilation. Dilation is performed by applying max pooling to the target masks (kernel size $k = 7$ for Landsat). For this experiment, model training and validation is conducted on dilated targets, but evaluation is performed on the original, undilated targets.

Each experiment (architecture, covariate set, pipeline stage) was defined by YAML configuration files specifying model hyperparameters, data paths, loss weights, and training settings for reproducibility.

Table 3: Summary of experimental configurations evaluated.

Exp.	**Model**	**Pipeline**	**Exp Config.**	**Purpose**
E1	HAUNet	Direct	Full (with pyramid encoder, no dilation)	Primary configuration
E1a	HAUNet	Direct	No pyramid encoder	Ablation: pyramid branch contribution
E1b	HAUNet	Direct	No DEM	Ablation: DEM contribution
E1c	HAUNet	Direct	No land cover	Ablation: land cover contribution
E1d	HAUNet	Direct	Full, dilated target (k=7)	Ablation: target representation (dilated vs. default)
E2.1a	HAUNet	Cascaded (S1)	Full, no dilation	Direct vs. cascaded
E2.1b	HAUNet	Cascaded (S1)	Full, dilated targets (k=3)	Ablation: target representation
E2.2a	HAUNet	Cascaded (S2)	Best config. from E1, E2.1a input	Direct vs. cascaded
E2.2b	HAUNet	Cascaded (S2)	Best config. from E1, E2.1b input	Ablation: target representation
E3	Autoencoder	Direct	Full	Baseline comparison
E3d	Autoencoder	Direct	Full, dilated target	Baseline comparison, matched to E1d
E4	Autoencoder	Cascaded	Full, same config. as best E2	Baseline, cascaded pipeline

In addition to default configuration, the autoencoder was also trained using the same target representation as HAUNet's best-performing configuration (dilated target, Section 5).

4.6 Evaluation Metrics

For the Stage 1 cascaded pipeline, to align the model prediction dimensions (256×256) with the target (20×20), an adaptive spatial max-pooling was applied to the output logits before calculating loss and metrics. All reported quantitative metrics including IoU, Boundary F1, AUPRC, and Centroid Distance reflect performance evaluated on the coarse 20×20 target.

- **AUPRC (Area Under the Precision-Recall Curve):** Measures the performance of the model across different thresholds, making it suitable for imbalanced datasets (Davis

and Goadrich, 2006):

$$\text{AUPRC} = \sum_k (R_k - R_{k-1}) P_k \tag{8}$$

where P_k and R_k represent precision and recall evaluated at the k-th threshold operating point. This is selected as the primary metric for evaluation as it is threshold-independent and less sensitive to extreme class imbalance. It captures the trade-off between precision and recall across all thresholds.

- **IOU (Intersection over Union):** Measures the overlap between predicted and ground truth. Given a binary prediction mask $\hat{Y} \in \{0,1\}^{H \times W}$ and binary target mask $Y \in \{0,1\}^{H \times W}$:

$$\text{IoU}(\hat{Y}, Y) = \frac{|\hat{Y} \cap Y|}{|\hat{Y} \cup Y|} = \frac{\sum_{i,j} \hat{Y}_{ij} Y_{ij}}{\sum_{i,j} \left[\hat{Y}_{ij} + Y_{ij} - \hat{Y}_{ij} Y_{ij} \right]} \tag{9}$$

- **Boundary F1:** Restricts precision and recall to fire-perimeter only, obtained by morphological erosion of the predicted and ground truth masks. This isolates edge/shape accuracy from interior-region, which pixel-level F1 can mask:

$$\text{F1}_b = 2 \cdot \frac{\text{Prec}_b \cdot \text{Rec}_b}{\text{Prec}_b + \text{Rec}_b + \epsilon} \tag{10}$$

where Prec_b and Rec_b are precision and recall computed only over boundary pixels of the predicted and ground truth masks.

- **Precision and Recall:** Evaluates the model's ability to correctly identify fire pixels (precision) and capture all true fire pixels (recall) (Saito and Rehmsmeier, 2015):

$$\text{Precision} = \frac{TP}{TP + FP + \epsilon}, \quad \text{Recall} = \frac{TP}{TP + FN + \epsilon} \tag{11}$$

Where TP, FP, and FN represent true positive, false positive, and false negative counts across the spatial grid, and $\epsilon = 10^{-8}$ prevents division by zero.

- **IPSNR (Intersection Peak Signal-to-Noise Ratio):** Measures the quality of predicted outputs within the intersection of predicted and ground truth fire regions (Badhan et al., 2024):

$$IPSNR = \log_{10} \left(\frac{\max(val)}{IRMSE} \right) \tag{12}$$

$$IRMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \cdot I}{\sum_{i=1}^{n} I}} \quad \text{where } I = \begin{cases} 1, & y_i \cdot \hat{y}_i \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where y_i and $\hat{y}_i$ are the ground truth and predicted pixel values, and I is an indicator function that selects only pixels where both the prediction and ground truth are non-zero. This is used over standard PSNR because the extreme sparsity of fire pixels means that PSNR is dominated by background pixels.

For Boundary F1, IOU, Precision, Recall, and IPSNR, the evaluation is performed on post-processed outputs using Otsu's thresholding to binarise the predicted downscaled masks (Zhang and Hu, 2008). This calculates the variance between the foreground and background classes to determine an optimal threshold t^* that maximises the separation between them. Given class probabilities $\omega_0(t), \omega_1(t)$ and means $\mu_0(t), \mu_1(t)$ for pixels above/below threshold t, the variance is:

$$\sigma_B^2(t) = \omega_0(t)\,\omega_1(t)\,\left(\mu_0(t) - \mu_1(t)\right)^2, \qquad t^* = \arg\max_t\ \sigma_B^2(t) \tag{14}$$

Otsu threshold is calculated and applied per patch. Otsu is used over a fixed threshold (e.g. 0.5) because the extreme class imbalance means that the model may predict low probabilities for the positive class, and a fixed threshold could lead to poor separation.

Due to the extreme rarity of the positive class, absolute metric values are difficult to interpret without a reference point. The following baselines were used:

1. an all-background predictor, assigning zero probability to every pixel.

2. a constant-prevalence predictor, assigning every pixel the measured test-set fire-pixel prevalence.

2. is the theoretically expected value of AUPRC under a classifier with no discriminative ability. Model performance is reported alongside both baselines.

While target dilation were used for the training target to mitigate class imbalance during optimisation, all evaluation metrics were computed against the original, undilated binary target masks. This ensures reported performance reflects the model's ability to predict at native target resolution.

The best model checkpoint for final evaluation was selected based on the highest validation AUPRC. AUPRC evaluates across range of classification thresholds, making it threshold-independent and less sensitive to the extreme class imbalance.

5 Results

The evaluation results for the direct and cascaded pipelines are summarised in Table 4.

IPSNR is computed only over the union of predicted and ground-truth fire regions and is therefore undefined for samples with no overlap between prediction and target. Consequently, its mean is sensitive to the frequency with which a model achieves overlap, independent of prediction quality. Therefore mean IPSNR is calculated only over samples with nonzero IoU, and the number of contributing samples shown in parentheses. The IPSNR values are low across all configurations, likely due to the extreme sparsity of fire pixels, meaning that even with opverlap, the predicted and ground-truth regions are often small.

6 Discussions

6.1 Overall Performance

Across configurations, absolute performance is generally low (most < 0.1 AUPRC and IOU). But all exceed both the All-Zero and Prevalence-Rate baselines, meaning the models are learning the fire signal rather than predicting background. But this should be taken as evidence of a working model rather than deployment-ready model.

Direct numerical comparison with Badhan et al., 2024 is not meaningful given differences in target resolution (2km $\rightarrow$ 375m single-stage vs. this study's 2km $\rightarrow$ 30m direct and cascaded mapping), prediction target (continuous BT regression vs. binary fire segmentation), and evaluation method. However, their reported IOU values (0.03-0.24) shows that even a single smaller resolution jump and less severe class imbalance (they use 20% fire-coverage threshold, and smaller image 128×128) still results in modest overlap. This suggests that the extreme resolution gap and class imbalance in this study make the low absolute performance unsurprising.

Table 4: Model performance evaluation across feature ablations vs. fixed baselines. Precision, Recall, Boundary F1, and IPSNR are computed on Otsu-thresholded predictions. IPSNR is reported as the mean over samples with nonzero post-processed IoU, with the contributing sample count in parentheses. Bold indicates the best value for that metric among the shown ablations of the same architecture and target

Exp.	**AUPRC**	**IoU**	**Precision**	**Recall**	**Boundary F1**	**IPSNR (n)**
Direct (GOES->Landsat) - HAUNet						
E1: Full, No Target Dilation	0.0103	0.0021	0.0021	0.3403	0.0030	0.02 (59/121)
E1a: No Cov. Pyramid Enc.	0.0098	0.0043	0.0043	0.4605	0.0039	0.03 (73/121)
E1b: No DEM	0.0050	0.0020	0.0020	0.3100	0.0033	0.02 (56/121)
E1c: No Land Cover	0.0054	0.0026	0.0026	**0.5051**	0.0020	0.02 (77/121)
E1d: Dilated Target	**0.0187**	**0.0044**	**0.0044**	0.3487	**0.0041**	0.04 (64/121)
Cascaded (GOES->VIIRS->Landsat) - HAUNet						
E2.1a: Stage 1 (no dilation)	**0.1458**	**0.0651**	**0.0721**	0.3582	**0.0956**	0.62 (807/1564)
E2.1b: Stage 1 (dilated)	0.1408	0.0505	0.0516	**0.6145**	0.0563	0.33 (1123/1564)
E2.2a: Stage 2 (E2.1a input, dilated)	**0.0147**	**0.0108**	**0.0125**	0.1365	0.0099	0.21 (31/121)
E2.2b: Stage 2 (E2.1b input dilated)	0.0134	0.0089	0.0095	**0.2435**	**0.0104**	0.10 (49/121)
Direct (GOES->Landsat) - Autoencoder						
E3: Full, No Target Dilation	**0.0120**	**0.0064**	**0.0077**	0.2592	0.0050	0.07 (51/121)
E3d: Dilated Target	0.0078	0.0063	0.0067	**0.3467**	**0.0063**	0.05 (67/121)
Cascaded pipeline - Autoencoder						
E4.1: Stage 1 (no dilation)	0.1408	0.0505	0.0516	0.6145	0.0563	0.33 (1123/1564)
E4.2: Stage 2 (dilated targets)	0.0185	0.0070	0.0070	0.3211	0.0060	0.07 (57/121)
Baseline: All-Zero	0.0006	0.0000	0.0000	0.0000	N/A	N/A
Baseline: Prevalence-Rate	0.0006	0.0000	0.0000	0.0000	N/A	N/A

6.2 Cascaded vs. Direct Pipeline

Comparing best configurations, the cascaded model's benefit over direct depends on the metric. For HAUNet, Stage 2 (E2.2a: IoU 0.0108, AUPRC 0.0147) outperforms the best direct configuration (E1d: IoU 0.0044, AUPRC 0.0187) on IoU but not AUPRC. This suggests the intermediate stage may help the model produce spatially better localised predictions (reflected in IoU/overlap-based metrics) but not necessarily improve confidence. Stage 1 metrics are much higher than either Stage 2 or direct-pipeline numbers (e.g. E2.1a AUPRC 0.1458), but this is due to Stage 1 metric being evaluated at the VIIRS target resolution (20×20). So metrics are not directly comparable.

6.3 Precision/Recall imbalance and threhsolding

Across nearly every configuration, Recall exceeds Precision after Otsu thresholding (e.g. E1c: Recall 0.5051 vs. Precision 0.0026; E1d: Recall 0.3487 vs. Precision 0.0044). This indicates the models are over-predicting positive (fire) pixels relative to the true sparse fire extent. However, in context of insurance, over-predicting fire pixels may be preferable to under-predicting, as false negatives (missed fire pixels) could have more severe consequences than false positives (over-predicted fire pixels). It also raises a question about whether Otsu is well-suited to thresholding predictions, as it assumes a roughly bimodal (two peaks) intensity distribution. A fixed threshold or one derived from the validation PR curve may be worth comparing in future work.

6.4 Effect of Target Dilation

Dilating the target mask improves AUPRC and IoU for HAUNet's direct configuration (E1d vs. E1: AUPRC 0.0187 vs 0.0103, IoU 0.0044 vs 0.0021) but not for the Autoencoder (E3d vs. E3: AUPRC 0.0098 vs 0.0120, IoU 0.0057 vs 0.0064, both lower with dilation). This may be because dilation allows for smoothing/tolerance for models like HAUNet that uses attention mechanisms to capture fine-grained information, while Autoencoder may benefit less from this beauce their convolutional nature are already prone to smoothing and wider predictions. This is consistent with the higher recall.

6.5 Feature Ablations

Out of direct HAUNet ablations, removing DEM (E1b) produces weakest performance (AUPRC 0.0050, IoU 0.0020) compared to full model (E1: AUPRC 0.0103, IoU 0.0021). This suggests topography is an important information that contributes to the model's ability to predict fire fronts. Removing land cover (E1c) increases Recall (0.5051, the highest in this group) but lower Boundary F1 (0.0020, the lowest in the group), suggesting that land cover helps produce more spatially precise predictions. Removing the covariate pyramid encoder (E1a) gives intermediate results across most metrics, suggesting it contributes some but not as much as DEM.

6.6 HAUNet vs. Autoencoder direct downscaling

In direct downscaling, Autoencoder outperforms HAUNet across AUPRC (0.0120 vs 0.0103) and IoU (0.0064 vs 0.0021). One possibility is because HAUNet is designed for fine-grained cross-channel and spatial dependencies in natural-images, and may be less suited to extremely coarse input (GOES) and sparse target which carries less information for the model to exploit.

6.7 Limitations of Cascaded Pipeline

A limitation of the cascaded pipeline is the use of Stage 2's intermediate predictions. Because true matching VIIRS/Landsat pairs are unavailable, Stage 2 is trained using Stage 1's predicted

VIIRS-resolution masks rather than true VIIRS observations. A consequence is that any error in Stage 1 propagates into Stage 2's training, meaning Stage 2 also learns to downscale Stage 1's errors alongside fire information. While cascading may reduce error from the resolution gap, it introduces source of error from stage-to-stage prediction dependency.

7 Conclusions

Future work should explore the integration of temporal information to capture the dynamic evolution of wildfires and/or predict burn areas. Another could include integration of climate data, and/or predicting fire intensity as well. Additionally, further research could investigate the use of generative and transformer-based models to enhance downscaled outputs and improve the model's ability to generalise across different geographic regions and fire regimes. The project could be extended to experiment with different architectures at each stage of the cascaded pipeline, or explore multi-stage cascades with more than two stages to further reduce the resolution gap at each step.

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